

HOMEWORK 1 FOR MATH 2860 (ELEMENTARY DIFFERENTIAL EQUATIONS)

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Due date: Sep 8th at 10:00 EST in class

Question Q1.

- a) Construct a slope field of the differential equation

$$y' = -\frac{x}{y}$$

for $y = -3, -2, -1, 0, 1, 2, 3$ and your own choice of x -values.

- b) From your slope field, **sketch** one integral curve.

Question Q2.

Find the solution to the differential equation

$$y' = \frac{1}{x^2 + x}; y(0) = 1.$$

In other words, integrate the function y' .

Question Q3.

- a) Which of the following is the general solution to the differential equation

$$y' - 2y + e^x = 0.$$

- i) $y = ce^{2x} + x$
- ii) $y = ce^{2x} + e^{-x}$
- iii) $y = ce^x + e^{-x}$
- iv) $y = e^x - ce^{2x}$

- b) From your answer in (a), solve the IVP

$$y' - 2y + e^x = 0, y(0) = 2.$$